Dorst on Updating

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1 Technical background

We fix a set of worlds, W. (This could be the set of all possible worlds, or some modelling substitute.)

A **probability function** (**probability**) on W is a function $\pi : \mathcal{P}W \to [0,1]$ such that $\pi(W) = 1$ and $\pi(A \cup B) = \pi(A) + \pi(B)$ when $A \cup B = \varnothing$.

- Note that when π and ρ are probabilities and $x \in [0,1]$, $x\pi + (1-x)\rho$ is a probability. A set Z of probabilities is **convex** iff whenever $\pi \in Z$ and $\rho \in Z$ and $x \in [0,1]$, $x\pi + (1-x)\rho \in Z$; **biconvex** iff Z and its complement are both convex.

An X**-valued random variable** (X**-concept**) is a function from W to X. So, a **probability-function valued random variable** (**probability concept**) is a function that maps each $w \in W$ to a probability function on W.

A probability concept P is **introspective** iff $P_w(P = P_w) = 1$ for every $w \in W$.

When π is a probability and P is a probability concept:

- P **is biased about** q **for** π iff $\pi(q) \neq \mathbb{E}_\pi(P(q))$.
- π **Reflects** P iff for every convex Z such that $\pi(P \in Z) > 0$, $\pi(\cdot \mid P \in Z) \in Z$. When W is finite, this is equivalent to: for every ρ such that $\phi(P = \rho) > 0$, $\pi(\cdot \mid P = \rho) = \rho$.
- π **Values** P iff for every "decision problem" (finite set of real-valued variables), any random variable that maps each w to $X(w)$ where X is an element of $\mathcal{D}$ for which $\mathbb{E}_{P_w}(X)$ is maximal has at least as high an expectation in π as any element of $\mathcal{D}$. (π is happy to outsource any decision to P.)
  - equivalently: iff for every real-valued random variable X, $\mathbb{E}_\pi(X \mid \mathbb{E}_P(X) > t) > t$ (π **Totally Trusts** P).
  - equivalently: iff for every biconvex Z, $\pi(\cdot \mid P \in Z) \in Z$.

We extend this terminology to relations between two probability concepts:

- P **Reflects** P' ($\langle P, P' \rangle$ **is Clearly Bayesian**) iff P_w Reflects P' for all w. P **Values** P' ($\langle P, P' \rangle$ **is Ambiguously Bayesian**) iff P_w Values P' for all w. P' is **biased about** q for P iff P' is biased about q for P_w, for some w.

Key facts:

- If π Reflects P, then $\pi\{w \mid P_w(P = P_w) = 1\} = 1$.
- If π Reflects P then $\pi\{w \mid P_w \text{ Reflects } P\} = 1$.
- If π Reflects P then P is unbiased about every q for π.
- If π Reflects P, π Values P.
- If π Values P then $\pi\{w \mid P_w \text{ Values } P\} = 1$.
- If π Values P and P Reflects itself, then π Reflects P.
- If π Values P and P is unbiased about every q for π, π Reflects P.

By contrast, non-introspective probability concepts can Value themselves, and can be Valued by probability functions for which they are biased about certain propositions.

An example of these two possibilities: Dorst's model of *Fast Coins*:

$$\pi = (1/3 \quad 1/3 \quad 1/3)$$

$$P = \begin{pmatrix} h & s & t \\ \hline 2/3 & 1/6 & 1/6 \\ 1/4 & 1/2 & 1/4 \\ 1/6 & 1/6 & 2/3 \end{pmatrix}$$

$\mathbb{E}_\pi(P(\neg s)) = (1/3)(5/6) + (1/3)(1/2) + (1/3)(5/6) = 5/18 + 3/18 + 5/18 = 13/18 > 2/3 = \pi(\neg s)$. So π treats P as biased about $\neg s$. Nevertheless, π Totally Trusts P.

- Considered as a toy model of how someone's credences might be disposed to vary depending on the ratio of heads to tails in the video they see, this has a couple of odd features.
  - First, it's "deterministic", in the sense that worlds that agree on the relevant question also agree as regards the agent's credences. A more adequate model would presumably be such that for any two worlds w_1 and w_2, there's a world w_3 where $P_{w_3} = P_{w_1}$ but w_3 agrees with w_2 on the proportion of heads in the video.

- Second, the fact that $P_h(s) = P_h(t)$ is quite weird: intuitively, you'd expect someone seeing *mostly-heads* to be much more confident in *split* than in *mostly-tails*. Seems more realistic to move some of P_h from t over to s:

$$P^x = \begin{pmatrix} h & s & t \\ \hline 2/3 & 1/3 - x & x \\ 1/4 & 1/2 & 1/4 \\ x & 1/3 - x & 2/3 \end{pmatrix}$$

  But there are hard limits on how much we can move: given the choice of P_s, we must have $x/(1/3 - x) \geq 1/3$, i.e. $x \geq 1/12$, if we want any probability function to be able to Totally Trust P^x.[1]

[1] In general: for a probability concept to Totally Trust itself (and a fortiori, for any probability to Totally Trust it), the probability function P_w assigned to each world must be some weighted average of (a) $P_w(\cdot \mid P = P_w)$, and (b) other possible values of P. See Dorst. et. al. 2021

There is also a Value-like relation that only makes sense for pairs of probability concepts:

- $\langle P, P' \rangle$ is **locally** Ambiguously Bayesian iff for some partition Q, (a) $P'_w(\cdot \mid q) = P_w(\cdot \mid q)$ for every cell q of Q, and (b) the restriction of P to Q Values the restriction of P' to Q.

2 *Normative claims about Reflecting and Valuing*

Philosophers have made various normative claims involving the Reflecting relation; roughly speaking, Kevin tends to reject these claims (on the grounds that they entail implausible introspection claims), but to be sympathetic to weakened claims that substitute Valuing for Reflecting.

In the paper by van Fraassen that introduced the word 'reflection', the focus is on probability-concepts of the form *x's credences at t*.[2]

[2] We have to restrict our attention to worlds where x is probabilistically coherent at all relevant time.

Future Credence Reflection For all x, t, Δ, if x is rational at t, then *x's credence function at t* Reflects *x's credences at* $t + \Delta$.

It's natural to consider weakening this to:

Future Credence Valuing For all x, t, Δ, if x is rational at t, then *x's credence function at t* Values *x's credences at* $t + \Delta$.

However, some influential objections to Future Credence Reflection also apply to Future Credence Valuing:

- Irrationality objection: I shouldn't now have high credence in *My head is a gourd* conditional on *At* $t + \Delta$ *I will have high credence that my head is a gourd*, assuming I now have high credence in *I will have lost my mind by* $t + \Delta$ conditional on that proposition.

- Forgetfulness objection: even if x is rational at t and certain that x will be rational at $t+\Delta$, x might still have nonzero credence that x will forget things between t and $t+\Delta$. E.g.: I now have high credence that *I eat Raisin Bran for breakfast on 14 Feb 2023* even conditional on *On 14 Feb 2024 I will have low credence that I eat Raisin Bran for breakfast on 14 Feb 2023*.

- Arntzenius's objection:

  > There are two paths to Shangri La, the Path by the Mountains and the Path by the Sea. A fair coin will be tossed to determine which path you will take: if heads you go by the Mountains, if tails you go by the Sea. If you go by the Mountains, nothing strange will happen: while traveling you will see the Mountains, and even after you enter Shangri La, you will forever retain your memories of that Magnificent Journey. If you go by the Sea, . . . just as you enter Shangri La, your memory of this Beauteous Journey will be erased and be replaced by a memory of the Journey by the Mountains.

  While viewing the Mountains, you should not have $\geq$0.5 credence in *I enter by the Sea* conditional on *I will have $\geq$0.5 credence that I enter by the Sea*. (Even if you have credence 1 in *I will remain rational the whole time and forget nothing*.)

Kevin's normative claims about Valuing are pretty different in structure from Future Credence Valuing: they focus on normative probability-concepts along the lines of 'the credences x should have at t'.[3] I think Kevin wants to say something like the following:

Should-Valuing For all x, t, Δ: *the credences x should have at t* Value *the credences x should have at $t+\Delta$*.

What's the operative sense of 'should'? I think it's co-ordinate with 'rational', which Kevin says he'll use 'as a context-sensitive term meaning "conforming to ideal ——"' (p.7).[4] The picture seems to be that for every ideal, agent, and time, there's a unique probability function that's the one "warranted" by that ideal for that agent at that time; some but not all ideals are 'sensible'; and the positive claim about Valuing involves tacit universal quantification over *sensible* ideals. 'Sensible' is in turn glossed in two (prima facie) different ways: (a) a sensible cognitive process is one that we would wouldn't prefer to get rid of if we realized we were subject to it; (b) a sensible cognitive process is one for which there is a computational-level (evolutionary) explanation.[5]

- I have trouble with the assumption of uniqueness built into 'the credences x should have at t'. Presumably there are multiple credence functions that it's metaphysically possible for x to have at t while

[3] It is presupposed that for every agent x and time t there's a *unique* such credence function.

[4] 'I've been talking about P, the opinions warranted by some particular ideal' (p.24).

[5] I'm not sure how to go from these characterizations of *processes* to characterizations of *ideals*, understood as giving rise to appropriate interpretations of 'should'.

satisfying the ideal: what singles out one of them as 'the' credence function x 'should' have?[6]

- I don't know which facts about a person's *actual* credence-forming methods matter in identifying the credences they *should* have. Consider, e.g., someone whose credence-forming dispositions w.r.t. *Fast Coins* are not surefire: when they view the video, there are several different credence functions over h, s, t they could end up with, and which one they end up with on a given occasion is a matter of (objective) chance. On some particular occasion they got unlucky—although h was true, they ended up with a credence function that assigned highest credence to t. Do we say that they *should* have highest credence in t? Or that they should have highest credence in h, on the grounds that that's the best outcome for their belief-forming dispositions?[7]

Q: does *Should-Valuing* avoid the objections to *Future Credence Valuing* discussed above? It certainly avoids the irrationality objection. To avoid the forgetting objection, we could say that the credences someone should have are not subject to forgetting.[8] To avoid Arntzenius's objection, one could say that after you get to Shangri La you *should* be confident that you entered via the Mountains, though that doesn't seem very plausible.

[6] A possible answer appeals to counterfactuals, using the controversial Conditional Excluded Middle principle in their logic. Then we could identify the credences x "should" have at t with the credences x *would* have if they satisfied the ideal at t—or maybe the credences they would have had at t if they satisfied the ideal throughout their life?

[7] I have worries either way. . . .

[8] Thus presumably the credences I should have right now are vastly more opinionated about my childhood than the credences I actually have.

3 *Arguing for the normative importance of Valuing*

Kevin gives three arguments (p.9–10) in favour of (something in the vicinity of) *Should-Valuing*.

1. If you are a counterexample to Should-Valuing (and have the credences you should have), there are some "proper" accuracy-measures for which your expectation at t of the accuracy of the credence function you should have at $t + \Delta$ is less than your accuracy at t.
2. If you are a counterexample to Should-Valuing (and have the credences you should have), there are decision problems for which you'd prefer to bind yourself now rather than leaving the option to your ideal future self.
3. If you are a counterexample to Should-Valuing you are vulnerable to a *fixed-option Dutch Book*, where you are sure all along what choice you'll face and that maximizing EU at each time will lead to a sure loss.

Conclusion: an ideal that doesn't fit (?) *Should-Valuing* 'is not sensible: it doesn't give a reflectively stable, computational-level explanation of why a cognitive system would work in this way. After all, an always-available option is to *ignore* the visual input and stay with the prior....' (p.10).[9]

- In response: even though a process that will replace your current credence function with the value of some probability-concept you don't Value has some downsides, it can also have big benefits (compared to staying put). It can increase your expected accuracy about lots of questions, and increase your expected profit given many hypotheses about what options you'll face. If evolution makes available to us processes which increase the correlation between our credences and the world, it seems there could be a computational-level evolutionary story about why we stick with those processes even if they mean our pre-process credences don't Value our post-process credences; and we might ourselves regard the benefits as worth the costs.[10]

[9] Kevin also seems to endorse the converse claim: 'showing that a cognitive process satisfies Value *suffices* to give a reflectively-stable, computational-level explanation of why a cognitive system would update beliefs in that way' (p.11, my emphasis). This doesn't make sense to me—surely there are many Value-satisfying processes, such as a "process" that completely ignores your visual inputs even though you have well-functioning eyes, whose evolution would be completely mysterious and which we'd prefer to change if we found out we had them.

[10] It would be different if there was some method of taking some sensory/cognitive process that generates systematic failures of Valuing and replacing it with a new process that avoids such failures, is no more demanding to implement, and preserves all the accuracy- and decision-theoretic benefits of the old process. But no such method has been proposed.

4 *Unmarked clocks and rotational symmetries*

Williamson ('Improbable Knowing') introduces a widely-discussed case that presents a strong challenge to (even the $\Delta = 0$ case of) Should-Valuing.

The concern does not depend on anything Williamson's own suggestion about the shape of the credence function you should have.[11] It arises if, e.g. the credence function you should have is a *normal* distribution centered on the truth.[12]

Nor does it depend on the assumption that the question what credence function you should have supervenes on the position of the clock hand.[13]

Dorst gives a model of the case where the agent has a "guess" about clock-position, and should have credence 1 that the guess is exactly whatever it in fact is. But the idea that we have that kind of access to facts about our guesses conflicts with Williamson's very influential anti-luminosity argument.[14]

[11] or for Williamson: the evidential probability function

[12] Conjecture: it's enough if there are angles θ, ρ such that the PDF drops off less between 0 and θ than between ρ and $\rho + \theta$.

[13] We can avoid assuming anything about what it depends on by just conditioning everything on the symmetry orbit of the actual world: the set of worlds in which all factors (not just the position of the clock hand but whatever determines what credence function you should have) are derived from those in actual world by some rotation. See Williamson, 'The KK principle and rotational symmetry'.

[14] *Knowledge and its limits.*

5 *Descriptive and explanatory claims*

Kevin proposes that various real-world facts about bias and polarization can be explained by the prevalence of certain Value-friendly, but Reflection-unfriendly, belief-forming mechanisms, or approximations thereof.[15]

[15] In combination with the normative claims discussed above, this hypothesis defeats the common use of such facts to argue for pervasive human irrationality.

Problem: the relevant expected-bias facts include cases where we can have high confidence that some process will leave us with high credence in some proposition in which we now have middling credence. This involves a failure of our current credences to Value our future credences.

Solution: hypothesise that because of our processing limitations, we are prone to engage in updates that are merely *locally* Ambiguously Bayesian. A string of several *locally* Ambiguously Bayesian updates can fail to be even locally Ambiguously Bayesian, if they involve different questions Q. That's what going on in the relevant cases.

Q: this seems acceptable if the goal is just that of providing a "computational level" explanation of the relevant tendencies; but is it supposed to also make the tendencies "reflectively stable"? Once we realize that sequences of locally Ambiguously Bayesian updates can lead to such accumulations of bias, shouldn't we start looking out for corrective mechanisms?